Universidade do Minho
Departamento de Informática

Dados e Aprendizagem Automática
Ano letivo 2026/2027

Practical Exercise no. 6

Theme

Ensemble Learning

Exercise

Ensemble Learning is an ML technique that combines several models to create an optimal prediction model. There are several types of Ensemble Learning, being the most common *Max Voting (Majority Voting)*, *BAGGing (Boostrap AGGregating)*, *Boosting* and *Stacking*:

- **Max Voting** involves fitting several models, each of which makes a prediction and votes on each sample. Only the class with the most votes is included in the final prediction;
- **Bagging** involves adjusting decision trees to different samples from the same dataset and calculating the average of these predictions;
- **Boosting** involves sequentially adding ensemble models that correct the predictions made by previous models, producing a weighted average of the predictions;
- **Stacking** involves fitting different types of models to the same dataset and using another model to determine the optimal combinations of predictions.

Tasks

The practical exercise includes the following tasks:

T1. Follow and implement the steps presented in the lab notes to design the various ensemble learning models.

T2. Analyze and compare the results of each model in terms of their performance and predictions. List your conclusions.

T3. Vary the hyperparameters and combinations of the models and analyze how each variation influences the optimal prediction model.

T4. Using the same dataset, change the paradigm to predict a continuous variable (*output = ["Fare"]*). Implement and train several models: Decision Tree, Linear Regression, Random Forest and Ensemble Learning, as discussed in the lab class.

T5. Analyze and compare the results of each model in terms of their performance and predictions. List your conclusions.

T6. Follow and implement the steps presented in the theoretical notes to design some ensemble learning models within the practical group work datasets.

T7. Critically analyze the results obtained.